8	**(a)(i)**	A spontaneous radioactive decay is one whereby it is <u>not triggered by external stimuli</u> such as <u>chemical composition and physical environment</u> (temperature, pressure, electric fields, magnetic fields, luminosity, etc.)	B1 B1
	(a)(ii)	$^{238}_{94}Pu \rightarrow ^{234}_{92}U + ^{4}_{2}He$ correct symbols correct mass and atomic number	 B1 B1
	(a)(iii)1.	Conservation of mass-energy K.E of products = $(m_{Pu} - m_u - m_\alpha)\, c^2$ $\left(5.649\times10^{6}\right)\left(1.6\times10^{-19}\right) = \left(238.0496 - m_u - 4.0026\right)\left(1.66\times10^{-27}\right)\left(3\times10^{8}\right)^2$ m_u = 234.0409 u	 C1 C1 A1
	(a)(iii)2.	By the principle of conservation of momentum, $0 = m_u v_u + m_\alpha v_\alpha$ $\frac{v_\alpha}{v_u} = -\frac{m_u}{m_\alpha}$ $\frac{T_\alpha}{T_u} = \frac{\frac{1}{2}m_\alpha v_\alpha^2}{\frac{1}{2}m_u v_u^2} = \frac{m_\alpha}{m_u}\left(-\frac{m_u}{m_\alpha}\right)^2 = \frac{m_u}{m_\alpha} = \frac{234.0409}{4.0026} = 58.47$ = 58.5	 C1 C1 A1
	(a)(iii)3.	$T_\alpha + T_u$ = 5.649 MeV $T_\alpha\,(1 + \frac{1}{58.47})$ = 5.649 MeV T_α = 5.55 MeV	 C1 A1
	(b)(i)	α -particles are highly ionising due to their charge and relatively large mass. Due to its ionising ability, an α -particle has very short range as it loses its energy rather rapidly. γ -rays are very weakly ionising because they are electromagnetic waves and carry no charge and no rest mass. Hence, they are most penetrating.	B1 B1
	(b)(ii)	Alpha particles are highly ionizing and will not penetrate further than 10 cm of air. Thus emission of α -particles will not be detected.	A1

<table>
<tr><td></td><td>(b)(iii)</td><td>True count = observed count – background count

t / hour	True count-rate / min^{-1}
0	6,409
6.0	801

$$\frac{C'}{C_0} = \frac{801}{6409} = \left(\frac{1}{2}\right)^n$$
$$\lg\left(\frac{801}{6409}\right) = n \ \lg\frac{1}{2}$$
$$\Rightarrow n = 3.0$$
$$\Rightarrow 6.0\,\text{hrs} = 3.0\, t_{1/2}$$
$$\therefore\ t_{1/2} = 2.0\,\text{hrs}$$
</td><td>C1

C1

A1</td></tr>
<tr><td></td><td>(b)(iv)</td><td>Short-term effect: Ionizing radiation can cause immediate damage to human tissue and could be accompanied by radiation burns, radiation sickness, etc.

Long-term effect: Delayed effects such as cataracts, hair loss, leukaemia, cancer, etc. could appear years after the exposure. Hereditary defects may also occur in succeeding generations as a result of gene damage.</td><td>B1

B1</td></tr>
</table>